

8. (a) (i) State why sodium is stored in oil in the laboratory. [1]

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(ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

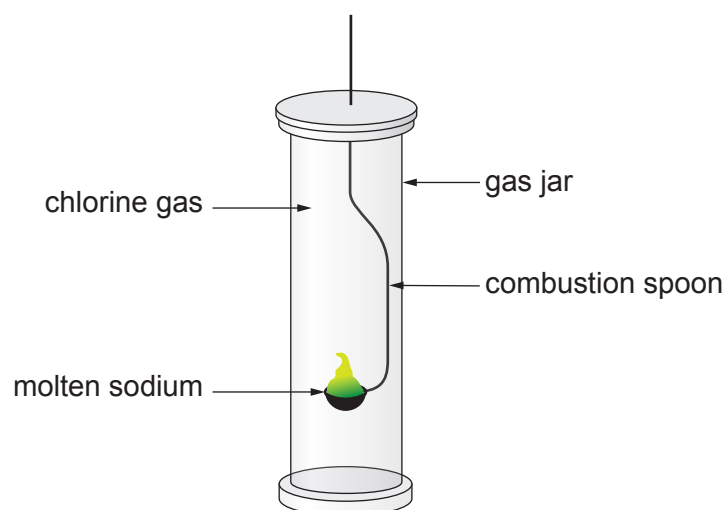
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(iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

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(b) The diagram shows the reaction of sodium with chlorine.



(i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

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(ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]



(c) The table shows some properties of Group 7 elements.

| Element  | Melting point (°C) | Boiling point (°C) | Reaction with hot iron |
|----------|--------------------|--------------------|------------------------|
| fluorine | −220               | −188               | explosive              |
| chlorine | −101               | −34                | very fast              |
| bromine  | −7                 | 59                 | quite fast             |
| iodine   | 114                |                    | slow                   |

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

−25 °C ☐      25 °C ☐      100 °C ☐      150 °C ☐

(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer. [2]

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